

RANSAC — From Greenhand to Expert

A complete, verified tutorial on RANdom SAmple Consensus

0 · How to read this tutorial

Part	Title	For you if...
I	Why RANSAC exists	you have never used it
II	The algorithm	you want to <i>understand</i> it, not just call it
III	The mathematics	you want to defend your parameter choices
IV	Build it yourself	you learn by writing code
V	The library toolbox	you have a deadline
VI	Geometric vision	you work with images or point clouds
VII	Failure modes	your RANSAC "works, except sometimes"
VIII	The variant zoo	you want the 2026 state of the art
IX	Neighbours & alternatives	you want to know when <i>not</i> to use it
X	Advanced topics	you fit multiple models, or want to learn the sampler
XI	Tuning playbook	you want a one-page cheat sheet
XII	Exercises	you want to check you really got it

Every code block in this document was executed before publication; every number in every table was computed, not recalled; every reference was checked against the primary source. Figure 1 is embedded below, and §4.4 contains the script that regenerates it.

1 · First things first: the name

RANSAC = RANdom SAmple Consensus.

Your message called it "*Randomized Sampling and Consensus*" — close in spirit, but not the official expansion, and the difference matters when you search the literature. The name comes from the title of the founding paper:

Martin A. Fischler and Robert C. Bolles, "**Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography**", *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, June 1981. DOI [10.1145/358669.358692](https://doi.org/10.1145/358669.358692) · free PDF at [SRI International](#)

Two further corrections to the framing in your question, because getting these straight early saves a lot of confusion later:

RANSAC is not a "math tool" in the sense of a theorem or a closed-form formula. It is a *randomized algorithm* — more precisely a **meta-algorithm** (a *paradigm*, in Fischler and Bolles's own word). It does not fit models itself. It wraps around a fitting routine that you supply, and makes that routine survive corrupted data. Swap in a line solver and you get robust line fitting; swap in a homography solver and you get image stitching. The wrapper never changes.

Its output is random. Two runs on identical input can return different answers. This is not a bug, it is the defining property, and §8.5 explains how to live with it.

Also worth knowing, since you will meet these strings in papers and code: the whole family of methods is called **SAC** (SAmple Consensus), which is why the variants are named MSAC, MLESAC, PROSAC, MAGSAC, VSAC, and so on. PCL's module is literally called `sample_consensus`.

Part I · Why RANSAC exists

2.1 · The breakdown-point problem

Classical estimators minimise a sum of squared residuals:

minimise $\sum r_i^2$ over all model parameters

This is optimal — provably, by Gauss–Markov — when every measurement is a true measurement corrupted only by zero-mean, finite-variance noise. It is *catastrophically* non-optimal when some measurements are not measurements of your model at all.

The distinction is the single most important idea in robust statistics:

	Noise	Outliers (gross errors)
Source	sensor jitter, quantisation, discretisation	wrong correspondence, wrong object, sensor failure, a second structure in the scene
Distribution	narrow, roughly Gaussian, centred on the model	arbitrary, unbounded, unmodelled
Magnitude	a fraction of a pixel / millimetre	anything at all
Least squares	handles it optimally	destroyed by a single one

The formal measure is the **breakdown point**: the smallest fraction of arbitrarily corrupted data that can drag an estimate arbitrarily far from the truth.

Estimator	Breakdown point
Ordinary least squares	0 % — one bad point with large leverage is enough
Huber M-estimator (fixed scale)	0 % — bounded influence in $\hat{y}$, none against high-leverage x
Least Median of Squares	50 % — the theoretical ceiling for a regression-equivariant estimator
RANSAC	>50 %, limited only by your compute budget and by ambiguity

That last row is the whole reason RANSAC has been cited more than 25,000 times and lives inside almost every camera, drone, robot, and 3-D scanner you will ever touch.

2.2 · A worked failure of least squares

Here is the canonical demonstration, with real numbers from a run you can reproduce (§4.4).

Generate 120 points on the line $y = 0.6x - 4$ with Gaussian noise $\sigma = 0.4$, then add 80 points drawn uniformly at random from the surrounding rectangle — a **40 % outlier rate**, which is entirely ordinary for image feature matching.

Method	Recovered slope	Recovered intercept	Slope error
Ground truth	0.600	−4.000	—
Least squares on all 200 points	0.319	−2.834	47 % low
RANSAC (10 trials, then refit on inliers)	0.598	−4.014	0.3 % low

Least squares did not degrade gracefully. It returned a line that fits nothing — pulled halfway towards the centroid of the outlier cloud, because squaring a residual of 20 contributes 2,500 times more to the objective than squaring a residual of 0.4. **The outliers outvote the signal by weight even though they are outnumbered by count.**

Panel A of Figure 1 shows this: the red least-squares line slicing through empty space while the green RANSAC line sits exactly on the black dashed ground truth.

RANSAC: what it does, how long it takes, and how to tune it

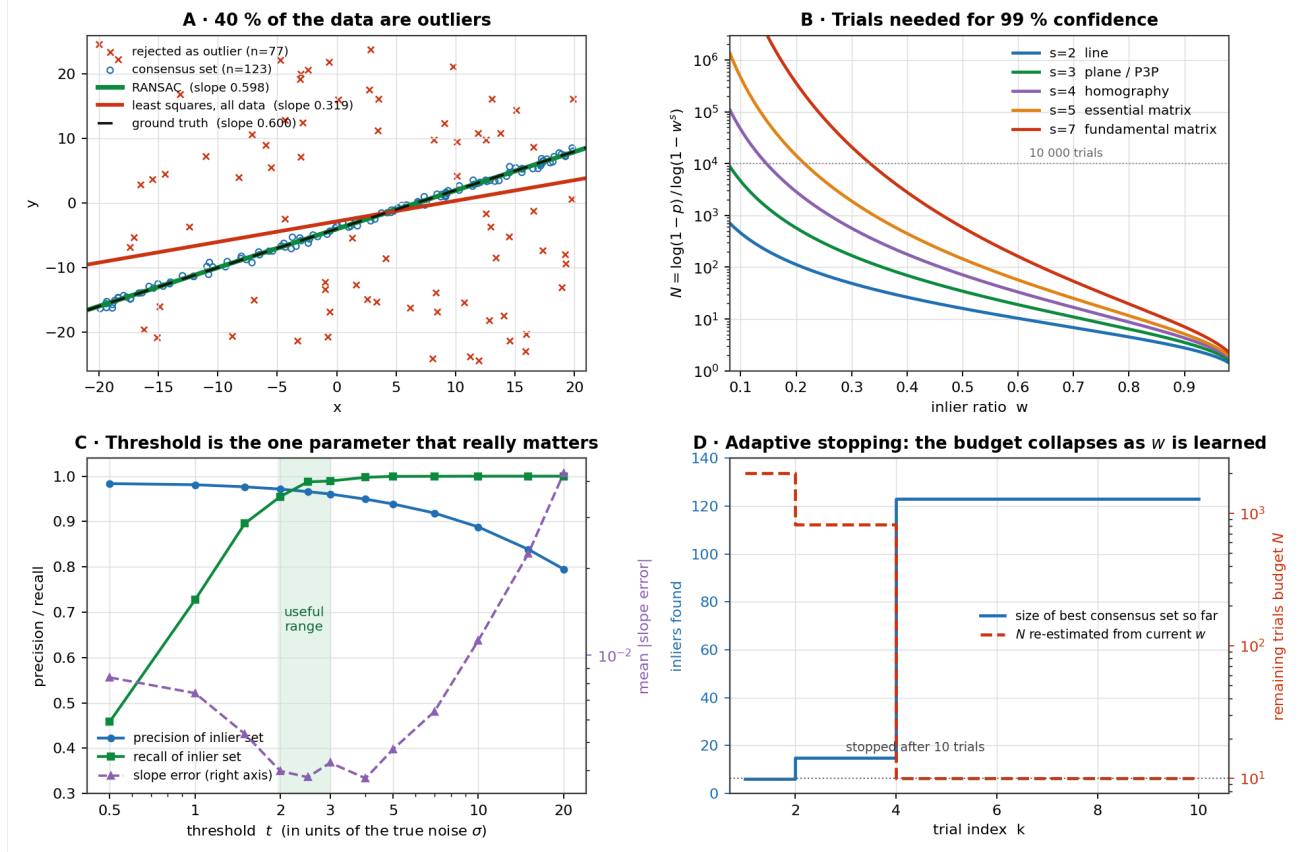


Figure 1. (A) The core idea, on data with 40 % outliers. (B) Required trials versus inlier ratio for the five most common minimal sample sizes. (C) The threshold sweep — precision, recall and accuracy versus threshold, averaged over 40 runs. (D) Adaptive termination in action: the trial budget collapses from 2,000 to 10 as the inlier ratio is discovered.

2.3 · The inversion of logic

The reason RANSAC feels strange at first is that it inverts the classical procedure.

	Classical fitting	RANSAC
Uses	all the data, once	a minimal subset, many times
Direction	data $\rightarrow$ model	model $\rightarrow$ data
Goal	minimise total error	maximise the number of points that agree
Rejects outliers by	down-weighting them	<i>excluding</i> them from a hypothesis, then testing

Fischler and Bolles's insight was this. You cannot identify the outliers before you know the model, and you cannot fit the model reliably before you remove the outliers — a chicken-and-egg deadlock. But if you pick just *two* points at random from 200, there is a reasonable chance (here, $0.6^2 = 36\%$) that **both happen to be inliers**. A model fitted to two clean points is already approximately correct, so most of the other inliers will fall close to it, and they can now be recognised. Repeat enough times and you will draw a clean pair almost surely.

So: **use uncontaminated small samples to reveal the inliers, and then use all the inliers to fit precisely**. The smaller the sample, the higher the chance it is clean — which is why RANSAC always uses the *minimal* sample size, not a comfortable one. Two points for a line, never three.

This also tells you immediately what RANSAC's cost structure is, and it is delightfully counter-intuitive: **the difficulty of a robust fitting problem is governed not by the number of data points, but by the minimal sample size and the inlier ratio.** A million points at 80 % inliers is easy — a handful of trials. Fifty points at 25 % inliers is easy too *if* you are fitting a line ($s = 2$, 72 trials), and close to hopeless if you are fitting a fundamental matrix ($s = 7$: over half a million trials, from twelve usable inliers).

Part II · The algorithm

3.1 · The four ingredients you must supply

RANSAC is a template with four holes. Fill them in and you have a robust estimator for your problem.

#	Ingredient	Symbol	Line fitting	Homography
1	Minimal solver — fits the model to the fewest points that determine it	s	2 points $\rightarrow$ line	4 correspondences $\rightarrow 3 \times 3$ H
2	Residual function — distance from a datum to a model, in meaningful units	$r(\theta, x)$	perpendicular distance	symmetric reprojection error, in pixels
3	Inlier threshold — how close counts as agreement	t	$\approx 2-3 \sigma$	1-4 px
4	Scoring rule — what makes one hypothesis better than another	—	count of inliers (better: MSAC, §5.4)	count of inliers (better: MSAC, §5.4)

Ingredient 3 is the only one that genuinely requires judgement, and it is where beginners lose most of their accuracy. §4.8 gives it a principled treatment and Panel C of Figure 1 shows the empirical cost of getting it wrong.

3.2 · Pseudocode

```
INPUT  X      : data, n items
       s      : minimal sample size
       t      : inlier threshold
       N      : maximum trials
       p      : desired confidence      (e.g. 0.99)

best_score  $\leftarrow -\infty$ 
best_inliers  $\leftarrow \emptyset$ 

for k = 1 ... N:

    # ---- 1. HYPOTHESEISE -----
    S  $\leftarrow$  s items drawn uniformly at random from X, without replacement
     $\theta \leftarrow$  minimal_solver(S)
    if  $\theta$  is undefined:                # degenerate sample
        continue

    # ---- 2. VERIFY -----
    r  $\leftarrow$  residuals( $\theta$ , X)          # all n items
    I  $\leftarrow$  { i :  $r_i < t$  }           # the consensus set

    # ---- 3. SCORE AND KEEP THE BEST -----
    if score(I, r) > best_score:
        best_score, best_inliers, best_ $\theta$   $\leftarrow$  score(I, r), I,  $\theta$ 

    # ---- 4. ADAPTIVE TERMINATION -----
    w  $\leftarrow$  |I| / n                      # inlier-ratio estimate
    N  $\leftarrow$  min( N, ceil( log(1-p) / log(1-w^s) ) )

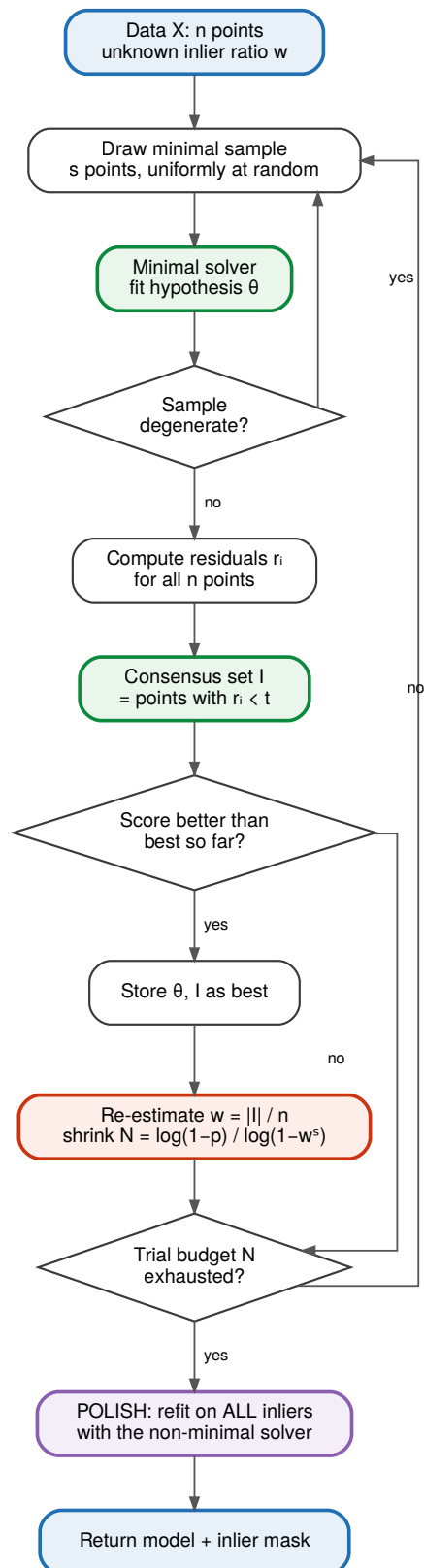
    # ---- 5. POLISH -----
     $\theta^* \leftarrow$  non_minimal_solver(X[best_inliers])    # least squares on ALL inliers

return  $\theta^*$ , best_inliers
```

Five stages: **hypothesise** $\rightarrow$ **verify** $\rightarrow$ **score** $\rightarrow$ **adapt** $\rightarrow$ **polish**. Every variant in Part VIII is a better idea about one of those five stages, and nothing else. Once you see the algorithm this way, the variant zoo stops being a zoo and becomes a table.

Step 5 is not optional. A model from a 2-point sample is *unbiased but noisy*: it used two measurements when 123 were available. Refitting on the full consensus set is what converts a rough hypothesis into a precise estimate. Measured over 200 runs of the line problem in §5.3, the polish step cuts the mean slope error from 0.0155 to 0.0047 — a **3.3× improvement, for free**; the gain grows with the number of inliers and with §5. Beginners who skip it wonder why RANSAC is "imprecise". RANSAC is not imprecise; a 2-point fit is.

3.3 · The loop, as a picture



3.4 · A micro-example you can check by hand

Six points, fitting a line, threshold $\tau = 1.0$:

i	point	on the line $y = x$?
1	(0, 0)	yes
2	(1, 1)	yes
3	(2, 2)	yes
4	(3, 3)	yes
5	(1, 9)	no — outlier
6	(2, -8)	no — outlier

Trial 1 draws points 5 and 6. The line through (1, 9) and (2, -8) is $y = -17x + 26$, i.e. $17x + y - 26 = 0$, normalised $(17x + y - 26)/\sqrt{290}$. Distances:

point	distance
(0,0)	$26/17.03 = \mathbf{1.53}$
(1,1)	$8/17.03 = 0.47 \checkmark$
(2,2)	$10/17.03 = 0.59 \checkmark$
(3,3)	$28/17.03 = \mathbf{1.64}$
(1,9)	0 $\checkmark$
(2,-8)	0 $\checkmark$

Consensus set size **4**. Note the trap: this catastrophically wrong model still collected four points, two of them genuine inliers that happen to lie near the bad line. A single trial proves nothing.

Trial 2 draws points 1 and 3. The line is $y = x$, i.e. $(x - y)/\sqrt{2} = 0$. Distances: points 1-4 are exactly 0 $\checkmark\checkmark\checkmark\checkmark$; point 5 is $8/1.414 = 5.66 \times$; point 6 is $10/1.414 = 7.07 \times$.

Consensus set size **4** again — tied! And this is exactly why the tie-break in the pseudocode matters. Trial 1's inliers have total residual $0.47 + 0.59 + 0 + 0 = 1.06$; trial 2's have total residual 0. Trial 2 wins on the tie-break, and the algorithm keeps the correct model.

Add a seventh true inlier at (4, 4) and the tie disappears: trial 2 would collect 5, trial 1 still 4. With realistic data, correct models win by a wide margin and ties never arise — but small problems are exactly where they do, and a scoring rule of "inlier count only" is genuinely ambiguous there. (This observation, taken seriously, is the origin of MSAC and MLESAC; see §9.1.)

Part III · The mathematics

This is the part that turns a user into an expert. Three questions have clean closed-form answers — how likely is a clean sample, how many trials do I need, how long will it really take — two have important corrections that most treatments omit (§4.5, §4.7), and two are the parameter choices you will actually be judged on (§4.8, §4.9).

4.1 · What is the probability that a single sample is clean?

Let w be the **inlier ratio** — the probability that a randomly chosen datum is an inlier — and let s be the minimal sample size. Drawing s items independently (see §4.5 for the without-replacement correction):

$$P(\text{sample is all-inlier}) = w^s$$

A clean sample is called **uncontaminated** or **all-inlier**. This one expression is the engine of every RANSAC cost analysis, and it carries a stark message: the probability decays *exponentially* in s . At $w = 0.5$, a 2-point sample is clean 25 % of the time, a 7-point sample 0.78 % of the time. **Every extra point in your minimal sample is exponentially expensive.** That is why the five-point algorithm for the essential matrix was a major result, and why people write papers about solvers that shave a single point off the sample.

4.2 · How many trials do I need? — the fundamental formula

The probability that *all* of N independent samples are contaminated is $(1 - w^s)^N$. We want the probability of at least one clean sample to be at least p (our confidence, conventionally 0.99):

$$1 - (1 - w^s)^N \geq p \iff (1 - w^s)^N \leq 1 - p$$

Take logarithms. Both $\log(1 - w^s)$ and $\log(1 - p)$ are negative, so dividing by the former flips the inequality:

$$N \geq \frac{\log(1 - p)}{\log(1 - w^s)}$$

```
log(1 - p)
N = ceil( ----- )    p = confidence (0.99)
log(1 - w^s)           w = inlier ratio
                      s = minimal sample size
```

Three properties worth internalising:

1. **n does not appear.** The number of data points is absent. Doubling your dataset does not change the number of trials by one — it only makes each trial more expensive to verify. This is RANSAC's most surprising and most useful property.
2. **N grows only logarithmically in $1/(1-p)$.** Going from 99 % to 99.99 % confidence merely doubles the trial count. Confidence is cheap; buy plenty.
3. **N grows like w^{-s} .** This is the wall. It is exponential in s and polynomial-of-high-degree in $1/w$, and it is why low-inlier-ratio problems with large minimal samples are the hard cases.

4.3 · The table every practitioner should know

Exact values of $N = \lceil \log(0.01) / \log(1 - w^s) \rceil$ for $p = 0.99$:

$w \downarrow s \rightarrow$	1	2 line	3 plane, P3P	4 homography	5 essential	6	7 fundamental	8 8-pt F
0.90	2	3	4	5	6	7	8	9
0.80	3	5	7	9	12	16	20	26
0.70	4	7	11	17	26	37	54	78
0.60	6	11	19	34	57	97	163	272
0.50	7	17	35	72	146	293	588	1 177
0.40	10	27	70	178	448	1 123	2 809	7 025
0.30	13	49	169	567	1 893	6 315	21 055	70 188
0.20	21	113	574	2 876	14 389	71 954	359 777	1 798 893
0.10	44	459	4 603	46 050	460 515	4 605 168	46 051 700	460 517 014

Read this table diagonally and the engineering reality jumps out. The top-left region is free. The bottom-right region is a wall you cannot climb with plain uniform sampling. You need either *fewer* samples (PROSAC, NAPSAC — they change which minimal sets you try), *cheaper* samples (SPRT, $T_{\{d,d\}}$ — these leave N untouched but slash the cost of each trial), or a higher w from a better matcher (Part VIII). Panel B of Figure 1 is this table as curves.

Two landmarks worth memorising: $w = 0.5, s = 4 \rightarrow 72$ trials (a homography from 50 %-correct matches is trivially cheap) and $w = 0.3, s = 7 \rightarrow 21,055$ trials (a fundamental matrix from 30 %-correct matches is a real computation).

4.4 · How long will it *actually* take?

N is a worst-case budget, not an expectation. The number of trials until the *first* clean sample is a geometric random variable with success probability $q = w^s$:

quantity	value	at $w=0.5, s=4$	at $w=0.4, s=7$
Expected trials to first clean sample	$1/q$	16	610
Standard deviation	$\sqrt{(1-q) / q}$	15.5	610
Budget N for $p = 0.99$	$\lceil \log(0.01) / \log(1-q) \rceil$	72	2 809

Notice that **the standard deviation is almost equal to the mean**. For a geometric distribution the coefficient of variation is $SD/mean = \sqrt{(1-q) / q}$, which is ≈ 1 whenever $q = w^s$ is small — so the spread of RANSAC's runtime is as large as the runtime itself. The practical consequences:

- **Never report a single timing.** Report the median and the 95th percentile over many runs.
- **Never budget for the mean** in a real-time system. Budget for N , or cap hard and detect failure.
- Because the mean is roughly $N/4.6$, RANSAC usually finishes far sooner than its worst case — which is exactly what adaptive termination (§4.6) exploits.

Here is the script behind the numbers above (Figure 1 is produced by `make_figs.py`, shipped alongside this document):

```
import math

def trials_needed(w, s, p=0.99):
    """Worst-case trial budget for confidence p."""
    return math.ceil(math.log(1.0 - p) / math.log(1.0 - w ** s))

def trial_stats(w, s, p=0.99):
    q = w ** s
    return {
        "P(clean sample)": q,
        "E[trials to first clean]": 1.0 / q,
        "SD[trials]": math.sqrt(1.0 - q) / q,
        "budget N": trials_needed(w, s, p),
    }

for w, s in [(0.5, 2), (0.5, 4), (0.4, 7), (0.3, 8)]:
    st = trial_stats(w, s)
    print(f"w={w} s={s}: N={st['budget N']:>6} "
          f"E={st['E[trials to first clean]']:>9.1f} "
          f"SD={st['SD[trials]']:>9.1f}")
```

```
w=0.5 s=2: N= 17 E= 4.0 SD= 3.5
w=0.5 s=4: N= 72 E= 16.0 SD= 15.5
w=0.4 s=7: N= 2809 E= 610.4 SD= 609.9
w=0.3 s=8: N= 70188 E= 15241.6 SD= 15241.1
```

4.5 · The correction almost everyone forgets: sampling without replacement

w^s assumes independent draws. Real implementations sample **without replacement** (you must not pick the same point twice — it would make the sample degenerate). The exact probability is hypergeometric: with $I = wn$ inliers among n points,

$$P(\text{clean}) = \frac{C(I, s)}{C(n, s)} = \prod_{j=0}^{s-1} \frac{I - j}{n - j}$$

which is **always smaller** than w^s . So the familiar formula is *optimistic*: it under-estimates the trials needed. A first-order expansion shows exactly what controls the gap:

$$\frac{P(\text{clean})}{w^s} \approx 1 - \frac{s(s-1)(1-w)}{2wn}$$

so the error scales with s^2 and with $1/(wn)$ — not with n/s , as is often assumed. Checked against exact values:

n	w	s	w^s	exact	exact / w^s	approximation above
1 000	0.40	7	0.001638	0.001587	0.969	0.969
200	0.50	4	0.062500	0.060620	0.970	0.970
200	0.50	7	0.007813	0.007009	0.897	0.895
100	0.60	4	0.129600	0.124358	0.960	0.960
50	0.50	7	0.007813	0.004813	0.616	0.580
20	0.50	4	0.062500	0.043344	0.693	0.700

Rule of thumb: the gap stays under 5 % when $n > 10 \cdot s(s-1)(1-w)/w$ — that is $n > 120$ for a homography at $w = 0.5$, but $n > 420$ for a 7-point fundamental matrix at the same w . Below that, the shortfall is real: estimating F from **40 correspondences at $w = 0.5$, the standard formula asks for 588 trials where 1,106 are genuinely needed — a 47 % under-estimate.** Multiply your budget by 2–3 for small n , or compute the hypergeometric probability directly:

```

from math import comb, log, ceil

def trials_needed_exact(n_inliers, n_total, s, p=0.99):
    """Trial budget accounting for sampling without replacement."""
    if n_inliers < s:
        return float("inf")
    q = comb(n_inliers, s) / comb(n_total, s)
    if q >= 1.0:
        return 1
    return ceil(log(1.0 - p) / log(1.0 - q))

print(trials_needed_exact(25, 50, 7)) # 955 (vs 588 from the w^s formula)

```

4.6 · Adaptive termination — never hard-code N

The formula needs w , which is precisely what you do not know. The fix, standard since Hartley and Zisserman's *Multiple View Geometry* (Algorithm 4.5), is beautifully simple: **start pessimistic, then re-estimate w from the best consensus set found so far and shrink N accordingly.**

```

N = float("inf") # or a large cap
k = 0
while k < N and k < max_trials:
    k += 1
    # ... hypothesise, verify ...
    if len(inliers) > len(best_inliers):
        best_inliers = inliers
        w = len(inliers) / n
        N = math.log(1 - p) / math.log(1 - w ** s) # monotonically shrinks

```

Panel D of Figure 1 shows the effect on the line-fitting problem. The budget sits at the 2,000 hard cap while only a 6-inlier model is known, drops to **817** at trial 2 when a 15-inlier model appears, and collapses to **10** at trial 4 the moment the 123-inlier model is found — at which point the loop exits. The run took **10 trials instead of 2,000**.

Three subtleties that separate the expert from the user:

1. **The estimate of w is biased upward**, because you are taking the *maximum* consensus size over many trials, and a maximum of noisy estimates over-estimates. So N is shrunk slightly too aggressively. In practice this is harmless; if you need to be careful, use the $N + \sqrt{(1-w^s)/w^s}$ safety margin of one standard deviation suggested by Hartley and Zisserman, or simply enlarge p .
2. **Always keep a hard `max_trials` cap.** With w genuinely tiny, N is astronomical, and an uncapped loop hangs. Every serious library has this cap: OpenCV's `maxIters` defaults to 2,000, scikit-learn's `max_trials` to 100.
3. **N can only decrease.** Never let a later, worse consensus set raise it again.

4.7 · What the guarantee does *not* say

This is the single most common expert-level misconception, so read it twice.

$N = \log(1-p)/\log(1-w^s)$ guarantees, with probability p , that **at least one sample consisting entirely of inliers was drawn**. It guarantees *nothing whatsoever* about the accuracy of the returned model, and nothing about that sample being *selected* as the best.

Three distinct things can go wrong even when the clean sample was drawn:

- **A clean sample can still be a bad sample.** Two inliers that happen to be adjacent give a line whose direction is wildly uncertain; four correspondences with any three of them collinear give a degenerate homography. Clean $\neq$ well-conditioned. This is why NeFSAC filters minimal samples for conditioning, and why LO-RANSAC's local optimisation helps so much.

- **A contaminated sample can win the scoring.** With many outliers and a loose threshold, a wrong model can collect more points than the right one — exactly the trap in §3.4. Panel C of Figure 1 quantifies it: at $t = 20\sigma$, inlier-set precision falls to 0.77 and the slope error grows 32-fold.
- **Degenerate configurations can produce a wrong model with genuinely high support.** The dominant plane problem in fundamental matrix estimation is the classic case (§8.1).

The practical upshot: **treat p as a budgeting device, not a correctness certificate.** Always validate the returned model independently — inlier count relative to expectation, residual distribution, geometric plausibility (§8.6).

4.8 · Choosing the threshold, properly

The threshold t is the one parameter with no safe default, and Panel C of Figure 1 shows why it deserves care. Averaged over 40 runs of the line problem with true noise $\sigma = 0.4$:

t (in units of σ)	inlier-set precision	inlier-set recall	mean slope error
0.5 σ	0.982	0.477	0.0064
1.0 σ	0.981	0.776	0.0047
2.0 σ	0.970	0.985	0.0032
3.0 σ	0.956	1.000	0.0033
5.0 σ	0.930	1.000	0.0049
10.0 σ	0.871	0.999	0.0199
20.0 σ	0.769	1.000	0.1031

The shape is a clean U: **too tight throws away real inliers** (recall 0.48 at 0.5 σ — you are fitting to half your data and the estimate gets noisy), **too loose admits outliers into the final least-squares refit** (precision 0.77 at 20 σ , and the slope error explodes by 32 $\times$). The optimum sits at **2–3 σ** , and it is broad — anywhere in 2–4 σ is fine. This is reassuring: you do not need to know σ precisely, only within a factor of about two.

Now the principled version. If inlier residuals are Gaussian with standard deviation σ , then the sum of squared residuals over the m degrees of freedom of the error is **chi-squared with m degrees of freedom**, where m is the *codimension* of the model — the number of independent directions in which a datum can deviate. Choosing t as the 95th percentile gives:

Model	Error measured as	Codim. m	t^2 / σ^2	t / σ
2-D line, 3-D plane	point-to-surface distance	1	3.841	1.96
Fundamental / essential matrix	point-to-epipolar-line distance	1	3.841	1.96
Homography, camera matrix	2-D reprojection error	2	5.991	2.45
Trifocal tensor	3-D transfer error	3	7.815	2.79

```
from scipy.stats import chi2
t_over_sigma = {m: float(chi2.ppf(0.95, m)) ** 0.5 for m in (1, 2, 3, 4)}
# {1: 1.960, 2: 2.448, 3: 2.795, 4: 3.080}
```

This is exactly the 2–3 σ band the experiment found, from first principles. Practical guidance:

- **Images, well-calibrated features (SIFT/SuperPoint + a good matcher):** $\sigma \approx 0.5$ –1 px, so $t = 1$ –3 px. OpenCV's own documentation recommends 1–10 px for `ransacReprojThreshold`.
- **Images, semantic or learned correspondences:** σ can be several pixels; $t = 4$ –10 px.
- **LiDAR / depth sensors:** use the datasheet noise figure. Typical indoor RGB-D plane fitting: $t = 0.01$ –0.02 m.

- **Unknown noise scale:** use **MAGSAC++** (§9.1), which marginalises over σ instead of fixing it. Benchmarks found it to be the only method whose optimal threshold was stable across datasets — which is exactly the property you want when you cannot tune.
- **Estimate σ from the data:** a robust scale estimate from the current inliers, $\hat{\sigma} = 1.4826 \cdot \text{MAD}(r)$, then $t = 2.5 \hat{\sigma}$, iterated 2–3 times, works well. This is the idea behind LO-RANSAC's inner loop. Note that scikit-learn's default is *cruder* than this — it uses the MAD of the target values y , not of the residuals, which is only a rough scale proxy. Set `residual_threshold` explicitly whenever you know your noise level.

4.9 • Computational complexity

Per trial you pay for one minimal solve plus n residual evaluations:

$$\text{Cost} = N (C_{\text{solve}} + n C_{\text{residual}}) + C_{\text{polish}}$$

For anything but tiny datasets, $n \cdot C_{\text{residual}}$ dominates completely — verification, not hypothesis generation, is the bottleneck. A fundamental-matrix RANSAC with $N = 20,000$ and $n = 2,000$ correspondences performs 40 million residual evaluations. This single observation launched an entire research line:

- **$T_{\{d,d\}}$ pre-test** (Chum & Matas, 2002): check d random points first (usually $d = 1$); if any fails, discard the hypothesis without touching the other $n - d$. Since the vast majority of hypotheses are bad, this eliminates most of the work.
- **SPRT** (Matas & Chum, 2005; Chum & Matas, 2008): Wald's sequential probability ratio test applied to verification — evaluate residuals one at a time and stop as soon as the evidence is decisive. Provably optimal in the expected-time sense, and the reason it is enabled for *every* `USAC_*` flag in OpenCV.
- **Preemptive RANSAC** (Nistér, 2003): generate a fixed pool of hypotheses and score them all in breadth-first order on a growing subset of data, pruning as you go. Bounded runtime, which is what a real-time system needs.

If you are writing your own, two cheap wins before you reach for any of these: **vectorise the residual computation** (one NumPy expression over all n points, never a Python loop) and **early-terminate scoring** once the running count cannot beat the best so far.

Part IV · Build it yourself

Nothing makes RANSAC click like implementing it. What follows is a complete, tested engine in about 120 lines of NumPy, plus three models. Every line has been executed; the output at the end of this section is the real output.

5.1 · The engine

Note the design: the engine knows nothing about geometry. It receives a `fit` function, a `residuals` function, and a sample size. That separation is what makes RANSAC a paradigm rather than an algorithm.

```

from __future__ import annotations
import math
from dataclasses import dataclass, field
from typing import Callable, Optional
import numpy as np

@dataclass
class RansacResult:
    model: object
    inliers: np.ndarray          # boolean mask, length n
    n_trials: int
    inlier_ratio: float
    residual_scale: float       # robust sigma estimate from the inliers
    converged: bool = field(default=True)

def ransac(
    data, fit, residuals, min_samples, threshold, *,
    max_trials=10_000, confidence=0.99, refit=None,
    score="msac", is_sample_valid=None, rng=None,
):
    """
    Robustly fit a model to `data` in the presence of outliers.

    data          (n, ...) array; row i is one datum.
    fit            fit(sample) -> model, or None if the sample is degenerate.
    residuals      residuals(model, data) -> (n,) non-negative errors.
    min_samples    s, the minimal sample size for `fit`.
    threshold      t, the inlier threshold, in the units of `residuals`.
    max_trials     hard cap; ALWAYS keep one.
    confidence     p, probability of drawing >= 1 uncontaminated sample.
    refit          non-minimal solver used to polish on the consensus set.
    score          "ransac" -> inlier count      (Fischler & Bolles 1981)
                  "msac"   -> truncated L2 cost  (Torr & Zisserman 2000)
    is_sample_valid optional cheap pre-check on the drawn sample.
    rng            seed or np.random.Generator, for reproducibility.
    """
    rng = np.random.default_rng(rng)
    data = np.asarray(data)
    n = len(data)
    if n < min_samples:
        raise ValueError(f"need >= {min_samples} data points, got {n}")

    t2 = threshold * threshold
    best_model, best_score = None, -np.inf
    best_inliers = np.zeros(n, dtype=bool)
    trials, budget = 0, max_trials

    while trials < min(budget, max_trials):
        trials += 1

        # ----- 1. HYPOTHESE -----
        idx = rng.choice(n, size=min_samples, replace=False)
        sample = data[idx]
        if is_sample_valid is not None and not is_sample_valid(sample):
            continue
        model = fit(sample)
        if model is None:                # degenerate configuration
            continue

        # ----- 2. VERIFY -----
        res = np.asarray(residuals(model, data), dtype=float)
        inliers = res < threshold
        n_in = int(inliers.sum())
        if n_in < min_samples:
            continue

        # ----- 3. SCORE -----
        if score == "ransac":
            # inlier count, total residual as tie-break
            current = n_in - 1e-9 * float(res[inliers].sum())
        else:
            # MSAC: truncated quadratic. Outliers each contribute t^2,
            # inliers contribute their squared residual. Lower is better,
            # so negate to keep "higher is better".

```

```

current = -float(np.minimum(res * res, t2).sum())

if current > best_score:
    best_model, best_inliers, best_score = model, inliers, current

# ----- 4. ADAPTIVE TERMINATION -----
w = n_in / n
denom = math.log(max(1e-12, 1.0 - w ** min_samples))
budget = min(budget,
              math.ceil(math.log(1.0 - confidence) / denom))

if best_model is None:
    return RansacResult(None, best_inliers, trials, 0.0,
                        float("nan"), converged=False)

# ----- 5. POLISH -----
if refit is not None and best_inliers.sum() >= min_samples:
    polished = refit(data[best_inliers])
    if polished is not None:
        res = np.asarray(residuals(polished, data), dtype=float)
        new_inliers = res < threshold
        # accept the polish only if it does not lose support
        if new_inliers.sum() >= best_inliers.sum():
            best_model, best_inliers = polished, new_inliers

final = np.asarray(residuals(best_model, data), dtype=float)
ir = final[best_inliers]
sigma = (1.4826 * float(np.median(np.abs(ir - np.median(ir)))))
        if ir.size else float("nan"))

return RansacResult(best_model, best_inliers, trials,
                    float(best_inliers.mean()), sigma)

```

Six details in there are the difference between a toy and something you would ship:

Detail	Why it matters
<code>fit</code> may return <code>None</code>	degenerate samples (coincident points, collinear triples) must be skipped, not crash
<code>min(budget, max_trials)</code>	adaptive termination and a hard cap; either alone is a bug
tie-break on total residual	breaks the §3.4 ambiguity; costs nothing
MSAC option	measurably more accurate — see §5.4
polish accepted only if support does not drop	guards the rare case where refitting drifts off the structure
<code>rng</code> parameter	reproducibility; §8.5 explains why you need this

5.2 · Three models

The pattern for every model is: a `fit` that works for **both** the minimal case and the over-determined case, and a `residuals` that returns a **geometrically meaningful** distance.

```

# ===== 2-D line:  $n \cdot x + c = 0$ ,  $|n| = 1$ ,  $s = 2$  =====
def fit_line(pts):
    """Total least squares — the minimal (2-point) and n-point cases at once."""
    pts = np.asarray(pts, float)
    centroid = pts.mean(axis=0)
    _, sv, vt = np.linalg.svd(pts - centroid)
    if sv[0] < 1e-12:
        return None
    normal = vt[-1]
    return normal, float(-normal @ centroid)

def line_residuals(model, pts):
    normal, c = model
    return np.abs(np.asarray(pts, float) @ normal + c)

def line_to_slope_intercept(model):
    (a, b), c = model
    if abs(b) < 1e-12:
        return float("inf"), float("nan")
    return -a / b, -c / b

# ===== 2-D circle:  $(cx, cy, r)$ ,  $s = 3$  =====
def fit_circle(pts):
    """Algebraic (Kasa) fit: linear least squares on  $x^2 + y^2 = ax + by + c$ ."""
    pts = np.asarray(pts, float)
    x, y = pts[:, 0], pts[:, 1]
    A = np.column_stack([x, y, np.ones(len(pts))])
    sol, *_ = np.linalg.lstsq(A, x * x + y * y, rcond=None)
    cx, cy = sol[0] / 2.0, sol[1] / 2.0
    disc = sol[2] + cx * cx + cy * cy
    if disc <= 0:
        return None
    return cx, cy, math.sqrt(disc)

def circle_residuals(model, pts):
    cx, cy, r = model
    pts = np.asarray(pts, float)
    return np.abs(np.hypot(pts[:, 0] - cx, pts[:, 1] - cy) - r)

# ===== 3-D plane:  $n \cdot x + d = 0$ ,  $s = 3$  =====
def fit_plane(pts):
    pts = np.asarray(pts, float)
    centroid = pts.mean(axis=0)
    _, sv, vt = np.linalg.svd(pts - centroid)
    # require genuine 2-D spread, else the sample is collinear/coincident
    if sv[1] < 1e-9 * max(sv[0], 1e-12):
        return None
    normal = vt[-1]
    return normal, float(-normal @ centroid)

def plane_residuals(model, pts):
    normal, d = model
    return np.abs(np.asarray(pts, float) @ normal + d)

```

Why `svd`, and not the two-point formula? Because the same function then serves as the `refit`. Using SVD on the mean-centred points computes the total-least-squares fit — minimising *perpendicular* distance — which is the correct objective when both coordinates are noisy. The naive `y = mx + b` least-squares fit minimises only vertical error and blows up for near-vertical lines. One function, both jobs, no special cases.

Why the `sv[1]` test for the plane and `sv[0]` for the line? Because degeneracy differs: a line needs two *distinct* points (the largest singular value must be non-zero), while a plane needs three *non-collinear* points (the *second* singular value must be non-zero). Getting these tests right is what prevents the mysterious `NaN` that plagues hand-rolled implementations.

5.3 • Running it

```
rng = np.random.default_rng(0)
M, B, SIG, NI, NO = 0.6, -4.0, 0.4, 120, 80    # 40 % outliers
x = rng.uniform(-20, 20, NI)
pts = np.vstack([
    np.column_stack([x, M * x + B + rng.normal(0, SIG, NI)]),
    np.column_stack([rng.uniform(-20, 20, NO), rng.uniform(-25, 25, NO)]),
])

r = ransac(pts, fit_line, line_residuals,
           min_samples=2, threshold=3 * SIG, refit=fit_line, rng=7)
slope, intercept = line_to_slope_intercept(r.model)
```

Actual measured output for the line (40 % outliers), circle (50 %), and plane (60 %):

```
LINE [ransac] slope=+0.5982 (true +0.6000) intercept=-4.0139 (true -4.0000) trials= 10
               inliers=123 precision=0.976 recall=1.000 sigma_hat=0.229
LINE [msac ] slope=+0.5982 (true +0.6000) intercept=-4.0139 (true -4.0000) trials= 10
               inliers=123 precision=0.976 recall=1.000 sigma_hat=0.229
LINE [OLS ] slope=+0.3192 (true +0.6000) intercept=-2.8339 (true -4.0000)
               <- destroyed by outliers

CIRCLE       centre=(+2.996,-1.981) r=5.000 (true (+3.0,-2.0) r=5.0)
               trials=36 inliers=103

PLANE        normal=[0.186 -0.281 0.942] d=-1.502 (true [0.188 -0.282 0.941] d=-1.500)
               angle_err=0.147 deg trials=65 inliers=124
```

A 60 %-outlier plane recovered to within **0.15 degrees in 65 trials**. Note also `sigma_hat = 0.229` against a true $\sigma = 0.4$: the robust scale estimate is biased *low*, because it is computed only over points that passed the threshold — a truncated distribution. Never use the inlier residual spread as an unbiased noise estimate.

5.4 • An experiment worth running yourself: does MSAC actually help?

Plain RANSAC scores a hypothesis by inlier *count*. MSAC (Torr & Zisserman, 2000) scores by a **truncated quadratic**: inliers contribute their squared residual, outliers each contribute the constant t^2 . Both have the same breakdown behaviour, but MSAC prefers a model whose inliers fit *tightly* over one that merely has as many.

On the easy problem above, at a well-chosen threshold, the two are indistinguishable — both give slope 0.5982. Which raises the fair question: why bother? Set the threshold badly and the answer appears. Averaged over 60 independent runs with `t = 4.0` (a sloppy **10 σ**):

Scoring rule	mean slope error	median slope error
"ransac" — inlier count	0.0382	0.0228
"msac" — truncated L2	0.0130	0.0068

MSAC is roughly 3× more accurate precisely when you have mis-set the threshold. Since mis-setting the threshold is the most common practical error, MSAC is close to a free lunch — which is why it is the *default* scoring function inside OpenCV's USAC framework, not an exotic option. Use it.

Part V · The library toolbox

Write it once to understand it; then use a library. The production implementations handle degeneracy, numerical conditioning, local optimisation, and SIMD verification far better than anything you will write in an afternoon.

6.1 · Which API for which job

Your task	Use this	Import
Robust linear regression	<code>RANSACRegressor</code>	<code>sklearn.linear_model</code>
Robust fit of a geometric primitive (line, circle, ellipse, similarity, affine, projective)	<code>ransac</code>	<code>skimage.measure</code>
Homography between two images	<code>cv2.findHomography</code>	<code>cv2</code>
Fundamental / essential matrix	<code>cv2.findFundamentalMat</code> , <code>cv2.findEssentialMat</code>	<code>cv2</code>
Camera pose from 3-D-2-D correspondences	<code>cv2.solvePnP</code>	<code>cv2</code>
Affine / rigid 2-D or 3-D transform	<code>cv2.estimateAffine2D</code> , <code>estimateAffinePartial2D</code> , <code>estimateAffine3D</code>	<code>cv2</code>
Plane in a point cloud	<code>segment_plane</code>	<code>open3d</code>
Planes, spheres, cylinders, cones in point clouds (C++)	<code>pcl::SACSegmentation</code>	PCL
State of the art two-view geometry	<code>USAC_MAGSAC</code> flag, or <code>pymagsac</code> / <code>pydegensac</code>	<code>cv2</code> , PyPI

6.2 · scikit-learn — robust regression

```
import numpy as np
from sklearn.linear_model import RANSACRegressor, LinearRegression

rng = np.random.default_rng(0)
X = rng.uniform(-20, 20, (200, 1))
y = 0.6 * X[:, 0] - 4.0 + rng.normal(0, 0.4, 200)
y[:80] = rng.uniform(-25, 25, 80) # corrupt 40 %

reg = RANSACRegressor(
    estimator=LinearRegression(), # NOTE: called `estimator`, not `base_estimator`
    min_samples=2,                # always set this explicitly
    residual_threshold=1.2,       # ~3 sigma; the parameter that matters
    max_trials=1000,
    stop_probability=0.99,
    loss="absolute_error",
    random_state=0,
)
reg.fit(X, y)

print(reg.estimator_.coef_, reg.estimator_.intercept_) # [0.596] -4.013
print(reg.n_trials_, reg.inlier_mask.sum())             # 10 127
```

Four traps, all of which bite beginners:

1. **The parameter is `estimator`, not `base_estimator`.** The old name was deprecated in scikit-learn 1.0 and removed in 1.2. Code and tutorials written before 2022 will not run.
2. **loss is "absolute_error",** not the old "absolute_loss" — also renamed.
3. **Always pass `min_samples` explicitly.** The default (`X.shape[1] + 1`) is only correct for `LinearRegression`; for any other estimator scikit-learn requires you to supply it.
4. **The default `residual_threshold` is the MAD of `y`** — a scale of the *target*, not of the residuals. That is a crude proxy and often far too loose. Set it yourself.

Useful extras: `is_data_valid` and `is_model_valid` callbacks let you reject samples or hypotheses cheaply — this is where you implement your own degeneracy tests.

6.3 · scikit-image — geometric primitives

```
import numpy as np
from skimage.measure import LineModelND, CircleModel, EllipseModel, ransac

data = np.column_stack([X[:, 0], y])
model, inliers = ransac(
    data, LineModelND,
    min_samples=2, residual_threshold=1.2, max_trials=1000, rng=0,
)
print(model.origin, model.direction)      # [2.33 -2.64] [0.859 0.512]
print(model.direction[1] / model.direction[0]) # 0.5968 -> the slope
print(inliers.sum())                      # 128
```

Note the **API change in scikit-image 0.26**: `model.params` is deprecated in favour of named attributes (`origin` / `direction` for `LineModelND`), and the internal fitting entry point moved from `Model.estimate()` to `Model.from_estimate()`. Code written against 0.25 or earlier will emit `FutureWarning` and will break in 2.2.

For image alignment, `skimage.transform` supplies the model classes directly — `EuclideanTransform`, `SimilarityTransform`, `AffineTransform`, `ProjectiveTransform` — and they plug straight into `ransac` with `min_samples` of 2, 2, 3, 4 respectively.

6.4 · OpenCV — the workhorse, and its modern flags

Since **OpenCV 4.5.0**, `calib3d` contains a full re-implementation of the USAC framework (contributed by Maksym Ivashechkin as a Google Summer of Code 2020 project). These flags are accepted anywhere the old `cv2.RANSAC` was, and they are substantially better:

Flag	What it actually is
<code>cv2.RANSAC</code>	the legacy implementation — still the default , and the weakest option
<code>cv2.LMEDS</code>	Least Median of Squares; no threshold needed, but hard-capped at 50 % outliers
<code>cv2.RHO</code>	PROSAC-style, for <code>findHomography</code> only
<code>cv2.USAC_DEFAULT</code>	LO-RANSAC + SPRT
<code>cv2.USAC_PARALLEL</code>	LO-RANSAC, multi-threaded
<code>cv2.USAC_FAST</code>	LO-RANSAC with fewer local-optimisation iterations
<code>cv2.USAC_ACCURATE</code>	GC-RANSAC (graph-cut local optimisation)
<code>cv2.USAC_PROSAC</code>	PROSAC sampling — input must be pre-sorted by match quality
<code>cv2.USAC_MAGSAC</code>	MAGSAC++ — the threshold-insensitive one
<code>cv2.USAC_FM_8PTS</code>	LO-RANSAC with the linear 8-point solver; fundamental matrix only

Every `USAC_*` flag uses SPRT verification and finishes with a non-minimal refit on all inliers. OpenCV's own benchmark study concluded that all of the new flags beat the legacy implementation, that `USAC_FAST` is never the right choice, and that `USAC_MAGSAC` is the only method whose optimal threshold was consistent across every dataset tested — the property that matters most when you cannot tune per-scene.

```

import cv2

# Homography. Defaults you should know: maxIters=2000, confidence=0.995.
H, mask = cv2.findHomography(
    src_pts, dst_pts,          # (N,1,2) float32
    method=cv2.USAC_MAGSAC,
    ransacReprojThreshold=3.0,  # pixels
    maxIters=10_000,
    confidence=0.9999,
)

# Fundamental matrix (s = 7)
F, mask = cv2.findFundamentalMat(p1, p2, cv2.USAC_MAGSAC,
    ransacReprojThreshold=1.0,
    confidence=0.9999, maxIters=10_000)

# Essential matrix (s = 5, needs calibration K)
E, mask = cv2.findEssentialMat(p1, p2, K, method=cv2.USAC_MAGSAC,
    prob=0.9999, threshold=1.0)

# Absolute pose from 3-D -> 2-D correspondences
ok, rvec, tvec, inliers = cv2.solvePnP(
    object_points, image_points, K, dist_coeffs,
    reprojectionError=3.0, confidence=0.9999,
    iterationsCount=10_000, flags=cv2.SOLVEPNP_EPNP,
)

# 2-D affine (s = 3) and rigid+scale (s = 2)
Maff, inl = cv2.estimateAffine2D(p1, p2, method=cv2.RANSAC, ransacReprojThreshold=3.0)
Mrig, inl = cv2.estimateAffinePartial2D(p1, p2, method=cv2.RANSAC, ransacReprojThreshold=3.0)

```

It is worth knowing which residual each estimator actually minimises, since that is what your threshold is measured in: **reprojection distance** for affine, homography and projection matrices (symmetric reprojection is also available for `H`), **Sampson distance** for the fundamental matrix, and **symmetric geometric distance** for the essential matrix.

Three OpenCV-specific gotchas:

- **method=0 means plain least squares**, with no robustness at all. It is easy to pass by accident, and the failure is silent. In the experiment in §7.2 it produced a 452-pixel error where RANSAC produced 0.14.
- **The mask is uint8, shaped (N, 1)**. Use `mask.ravel().astype(bool)` before indexing.
- **USAC solvers are deterministic by design** — identical input gives identical output. To get a different sample sequence, change `randomGeneratorState`. The exception is `USAC_PARALLEL`: OpenCV's documentation notes that with default options parallel RANSAC is *not* deterministic, because the result depends on how the threads interleave. For the legacy `cv2.RANSAC`, use `cv2.setRNGSeed`.

6.5 · Point clouds — Open3D and PCL

```

# ----- Open3D (Python) -----
import open3d as o3d

pcd = o3d.io.read_point_cloud("scene.ply")
plane_model, inlier_idx = pcd.segment_plane(
    distance_threshold=0.01,  # metres - your sensor's noise scale
    ransac_n=3,              # minimal sample for a plane
    num_iterations=1000,
)
a, b, c, d = plane_model      # ax + by + cz + d = 0
ground = pcd.select_by_index(inlier_idx)
objects = pcd.select_by_index(inlier_idx, invert=True)

```

```
// ----- PCL (C++) -----
#include <pcl/sample_consensus/method_types.h>
#include <pcl/sample_consensus/model_types.h>
#include <pcl/segmentation/sac_segmentation.h>

pcl::SACSegmentation<pcl::PointXYZ> seg;
seg.setOptimizeCoefficients(true); // = the polish step of §3.2
seg.setModelType(pcl::SACMODEL_PLANE); // also LINE, CIRCLE2D, SPHERE,
// CYLINDER, CONE, NORMAL_PLANE, ...
seg.setMethodType(pcl::SAC_RANSAC); // also SAC_LMEDS, SAC_MSAC,
// SAC_RRANSAC, SAC_RMSAC,
// SAC_MLESAC, SAC_PROSAC

seg.setMaxIterations(1000);
seg.setDistanceThreshold(0.01);
seg.setInputCloud(cloud);
seg.segment(*inliers, *coefficients);
```

PCL is the richest model library of the lot — it ships parallel-plane, perpendicular-plane, and normal-constrained variants, which let you inject prior knowledge ("the floor is roughly horizontal") directly into the hypothesis test. Doing so raises the effective inlier ratio and can cut the trial count by orders of magnitude. **Constraining the model is always cheaper than increasing N .**

Part VI · RANSAC in geometric vision

This is RANSAC's native habitat. Feature matching produces correspondences that are wrong 20–80 % of the time, and no amount of better descriptors removes the need for a robust estimator.

7.1 · Minimal sample sizes — the table to bookmark

s determines your cost (§4.3), so knowing it for your problem is the difference between 72 trials and 70,188.

Model	s	Solutions	Notes
2-D line	2	1	
2-D circle	3	1	collinear triples are degenerate
3-D plane	3	1	collinear triples are degenerate
3-D sphere	4	1	coplanar quadruples are degenerate
2-D translation	1	1	
2-D similarity (rot + scale + trans)	2	1	
2-D affine	3	1	collinear triples are degenerate
2-D homography	4	1	no 3 of the 4 may be collinear
3-D rigid (3-D $\leftrightarrow$ 3-D, Horn / Kabsch)	3	1	
Essential matrix (calibrated)	5	up to 10	Nistér's five-point algorithm
Fundamental matrix (7-point)	7	up to 3	enforces $\det F = 0$
Fundamental matrix (8-point, linear)	8	1	simpler, needs Hartley normalisation
P3P — pose from 3-D $\leftrightarrow$ 2-D, calibrated	3	up to 4	a 4th point disambiguates
Camera matrix P (uncalibrated pose)	6	1	
Trifocal tensor	6	up to 3	

Two lessons hide in this table. First, **calibration pays for itself**: knowing K lets you estimate an essential matrix from 5 points instead of a fundamental matrix from 7, cutting the trials at $w = 0.4$ from 2,809 to 448 — a 6× speed-up for free. Second, **the 7-point solver is preferable to the 8-point one inside RANSAC** despite being more complex, because the exponent in w^{-s} matters more than the cost per solve.

7.2 · Measured: accuracy versus inlier ratio

600 synthetic correspondences under a known homography, Gaussian noise $\sigma = 0.5$ px, varying outlier fraction. Median mean-transfer-error over a 48×48 grid across 25 runs, $t = 3$ px, `maxIters` = 100,000, `confidence` = 0.999:

Inlier ratio w	Least squares (method=0)	cv2.RANSAC	cv2.USAC_MAGSAC	RANSAC time
0.90	19.7 px	0.059 px	0.054 px	1.2 ms
0.70	59.8 px	0.061 px	0.065 px	2.0 ms
0.50	305.4 px	0.070 px	0.071 px	5.2 ms
0.35	252.5 px	0.085 px	0.084 px	19.1 ms
0.25	393.7 px	0.096 px	0.101 px	70.9 ms
0.15	452.6 px	0.136 px	0.142 px	544.3 ms

This is the single most instructive experiment in the tutorial, and it contains two separate messages that beginners routinely merge:

1. **Accuracy is astonishingly robust.** From 10 % outliers to 85 % outliers, the error grows only from 0.06 px to 0.14 px — a factor of 2.3 — while least squares degrades by a factor of 23 into total nonsense. RANSAC does not *degrade* as outliers increase; it either works or it doesn't.
2. **Runtime is not robust at all.** The same sweep costs 1.2 ms at `w = 0.9` and 544 ms at `w = 0.15`: **445× slower**. This is `w-5` from §4.2 showing up in a wall-clock measurement.

So: **if RANSAC is too slow, do not tune RANSAC — raise the inlier ratio.** A better matcher, a ratio test, mutual-nearest-neighbour filtering, or a motion prior will each buy you more than any amount of parameter fiddling. Going from `w = 0.25` to `w = 0.5` is a 14× speed-up; no implementation detail comes close.

7.3 · A complete, verified image-matching pipeline

```
import numpy as np
import cv2

def align_images(img1, img2, ratio=0.75, thresh=3.0):
    """Estimate the homography mapping img1 -> img2. Returns (H, stats)."""
    # ---- 1. detect and describe -----
    sift = cv2.SIFT_create()
    k1, d1 = sift.detectAndCompute(img1, None)
    k2, d2 = sift.detectAndCompute(img2, None)
    if d1 is None or d2 is None or len(k1) < 4 or len(k2) < 4:
        return None, {"reason": "too few features"}

    # ---- 2. match, with Lowe's ratio test -----
    # This is your cheapest possible increase of w. Do not skip it.
    knn = cv2.BFMatcher().knnMatch(d1, d2, k=2)
    good = [m for m, n in knn if m.distance < ratio * n.distance]
    if len(good) < 4:
        return None, {"reason": "too few matches", "n_matches": len(good)}

    src = np.float32([k1[m.queryIdx].pt for m in good]).reshape(-1, 1, 2)
    dst = np.float32([k2[m.trainIdx].pt for m in good]).reshape(-1, 1, 2)

    # ---- 3. robust estimation -----
    H, mask = cv2.findHomography(
        src, dst, cv2.USAC_MAGSAC,
        ransacReprojThreshold=thresh, maxIters=10_000, confidence=0.9999,
    )
    if H is None:
        return None, {"reason": "estimation failed", "n_matches": len(good)}

    inliers = mask.ravel().astype(bool)
    w = inliers.mean()

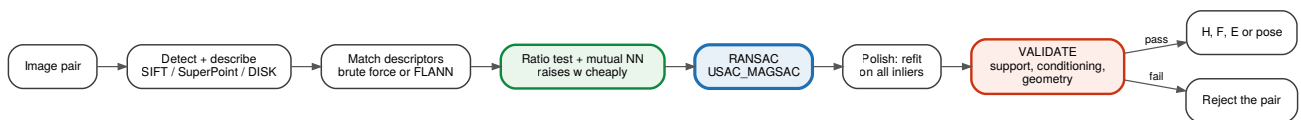
    # ---- 4. VALIDATE. Never trust the output blindly (§4.7, §8.6) --
    stats = {"n_matches": len(good), "n_inliers": int(inliers.sum()),
            "inlier_ratio": float(w)}
    if inliers.sum() < 15 or w < 0.15:
        return None, {"reason": "insufficient support"}
    if abs(np.linalg.det(H)) < 1e-8:
        return None, {"reason": "degenerate homography"}

    return H, stats
```

Measured on a synthetic pair with a known ground-truth homography (3,586 and 2,803 SIFT keypoints, 1,752 matches surviving the ratio test, i.e. `w ≈ 0.91`):

Method	Inliers	Mean transfer error	Time
<code>method=0</code> (least squares)	—	21.939 px	—
<code>cv2.RANSAC</code>	1599 / 1752	0.078 px	1.9 ms
<code>cv2.LMEDS</code>	1598 / 1752	0.072 px	9.6 ms
<code>cv2.RHO</code>	1142 / 1752	0.184 px	1.4 ms
<code>cv2.USAC_DEFAULT</code>	1599 / 1752	0.074 px	8.5 ms
<code>cv2.USAC_ACCURATE</code>	1599 / 1752	0.074 px	14.7 ms
<code>cv2.USAC_MAGSAC</code>	1598 / 1752	0.062 px	2.1 ms

Even at a comfortable 91 % inlier ratio, **the 9 % of bad matches inflate the least-squares error by a factor of 280**. There is no inlier ratio at which you can safely skip the robust estimator.



Every downstream system you have heard of is this diagram with more engineering: Structure from Motion (COLMAP), visual SLAM (ORB-SLAM), panorama stitching, visual localisation, and the loop closure in every LiDAR odometry stack.

Part VII · Failure modes, and how to detect them

RANSAC fails in a specific, small set of ways. Learn to recognise them and you will diagnose in minutes what otherwise costs days.

8.1 · Degeneracy — the failure that produces *confident* nonsense

A **degenerate** sample is one from which the model is not uniquely determined. Three collinear points do not determine a plane; four correspondences with any three of them collinear do not determine a homography. The solver either returns garbage or silently returns *something*, and that something can attract a large consensus set.

The notorious case is the **dominant plane** in fundamental-matrix estimation. If most of your correspondences lie on a single physical plane — a building façade, a table top, a road surface — then those correspondences are related by a homography H , and *every* fundamental matrix of the form $F = [e']_x H$ — a two-parameter family, one for each choice of epipole e' — explains all of them equally well. The epipolar geometry is genuinely under-determined by the planar points alone, so the answer is decided by the handful of off-plane points, which RANSAC's inlier count has no reason to prefer.

Measured, on a two-view scene where the fraction of points lying on one plane is swept. Error is the median Sampson distance evaluated **only on the off-plane points** — the ones that actually constrain F :

Fraction on the dominant plane	cv2.FM_RANSAC (legacy)	cv2.FM_LMEDS	cv2.USAC_FM_8PTS	cv2.USAC_MAGSAC
0.00	0.373 px	0.299 px	0.262 px	0.260 px
0.50	0.395 px	0.316 px	0.270 px	0.268 px
0.80	0.932 px	0.386 px	0.263 px	0.257 px
0.90	1.600 px	4.357 px	0.251 px	0.243 px
0.95	4.448 px	3.911 px	0.216 px	0.212 px

The legacy estimator degrades by **12×** and LMedS by as much as **15×**, while both USAC flags are completely flat — they are actually slightly *better* at 95 % because the surviving off-plane points are fitted more tightly. This is not luck. OpenCV's USAC framework carries an explicit degeneracy stage, and its documentation is specific: it applies DEGENSAC (Chum, Werner & Matas, 2005), which detects when **at least 5 of the 7 points in the minimal sample lie on the dominant plane** and recovers the correct model rather than accepting the degenerate one.

The lesson is blunt: on `findFundamentalMat`, do not use the default flag. Pass `cv2.USAC_MAGSAC` or `cv2.USAC_FM_8PTS`, or use the `pydegensac` package. This is the single highest-value one-line change in this entire tutorial.

Other degeneracies and their defences:

Degeneracy	Defence
Collinear / coincident points in the sample	return <code>None</code> from the solver (the <code>sv[0]</code> , <code>sv[1]</code> tests in §5.2). OpenCV's USAC runs a collinearity test for affine and homography samples
Planar scene for F	DEGENSAC / <code>USAC_*</code> flags; or estimate H instead and test which model the data prefer
Points behind the camera for F / E	the oriented epipolar constraint (Chum, Werner & Matas, 2004), applied inside USAC
Points on a conic for H	check the condition number of the design matrix
Pure rotation for E	there is no translation to recover — detect it and fall back to a rotation-only model
Quasi-degenerate data	QDEGSAC (Frahm & Pollefeys, 2006)

8.2 · Multiple structures — "pseudo-outliers"

If your data contain **two** valid structures, the points of structure B are outliers with respect to structure A — but they are not *random* outliers. They are organised, and organised outliers can form a consensus. Fischler and Bolles's framework assumes a single model; two models break the assumption.

Measured, on two crossing lines (100 points each, $\sigma = 0.2$), sweeping the threshold.

Contamination is the fraction of the returned consensus set that came from the *other* line:

Threshold	Mean consensus size (of 200)	Contamination from the other line	Mean recovered slope (true 1.000)	Runs returning a true bridging line
0.5	104.3	0.041	1.002 ± 0.007	0 / 30
2.0	116.8	0.143	1.014 ± 0.080	0 / 30
6.0	146.2	0.316	1.194 ± 0.481	8 / 30

At a tight threshold RANSAC does the right thing: it recovers one line almost exactly (slope 1.002 ± 0.007) with a handful of stray points near the crossing. As the threshold loosens, two distinct things go wrong. First, the returned line is *pulled* towards the other structure — mean slope drifts to 1.194 with a standard deviation of 0.481, so individual runs are wildly scattered. Second, in **8 of 30 runs at $t = 6$** the answer is a genuine **bridging model**: a line that is neither of the two real lines but straddles both, and therefore collects more support than either. Because the consensus set is *larger*, RANSAC's own score actively prefers it.

This is why a big inlier count is not evidence of correctness. Defences:

- Keep the threshold tight; multi-structure data punish looseness far more than single-structure data do.
- Use **sequential RANSAC**: fit, remove inliers, refit on the remainder (§10.1). Simple and often sufficient.
- Use spatially-aware sampling (**NAPSAC**, **P-NAPSAC**) — samples drawn from a local neighbourhood are far more likely to belong to a single structure.
- Use a proper multi-model method: **Progressive-X**, or energy-minimisation-based multi-model fitting.

8.3 · Low inlier ratio

When w is small, N explodes as w^{-5} (§4.3, and $445\times$ measured in §7.2). Plain uniform sampling eventually becomes infeasible. In priority order:

1. **Raise w** . Better matcher, Lowe's ratio test, mutual nearest neighbours, a motion or gravity prior. This is always the biggest win.
2. **Reduce s** . Calibrate the camera and use the 5-point essential solver instead of the 7-point fundamental one ($2,809 \rightarrow 448$ trials at $w = 0.4$). Exploit affine correspondences or known gravity direction to reduce s further.
3. **Sample non-uniformly**. **PROSAC** draws from the highest-quality matches first and, on typical data, finds a good model in tens of samples where uniform sampling needs thousands.
4. **Verify faster**. SPRT and $T_{\{d,d\}}$ attack the $n \cdot C_{\text{residual}}$ term (§4.9).

8.4 · The threshold is wrong

Symptoms and cures, from Panel C of Figure 1:

Symptom	Likely cause	Fix
Very few inliers; unstable answer between runs	t far too small	raise t , or estimate σ from the data
Huge inlier count but visibly wrong model	t far too large (§8.2 bridging)	lower t
Works on one dataset, fails on the next	t tuned to one noise level	use MAGSAC++ , which marginalises over σ
Accuracy worse than expected despite good inliers	polish step missing, or t loose	add the refit; use MSAC scoring

8.5 • Non-determinism

RANSAC is randomised, so identical input can give different output. This causes real problems: non-reproducible bugs, flaky tests, and jittery estimates in a video stream.

```
# Reproducibility
rng = np.random.default_rng(42)      # your own implementation
cv2.setRNGSeed(42)                   # legacy cv2.RANSAC
# The USAC_* solvers are deterministic by construction: same input ->
# same output. Vary `randomGeneratorState` if you want a different draw.
```

For temporal stability in video, do not simply re-run RANSAC per frame. Either seed the sampler with the previous frame's inliers, or use the previous model as a prior and run RANSAC only when the support drops — the standard trick in visual odometry.

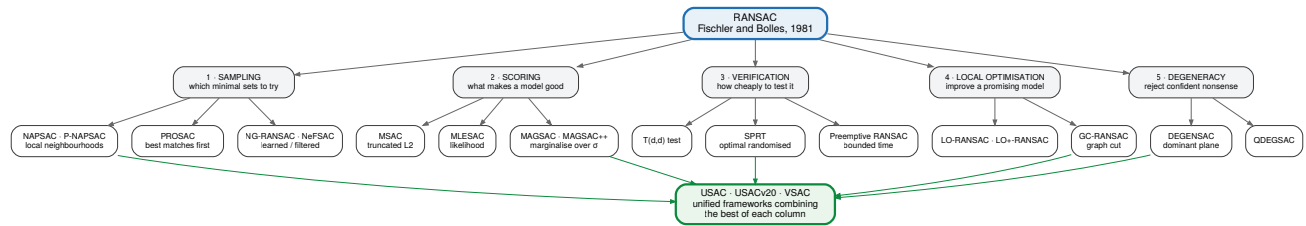
8.6 • Trusting the output — a validation checklist

Because the confidence p says nothing about correctness (§4.7), validate explicitly. Run through this list whenever a RANSAC result feeds anything important:

- ☐ **Absolute support.** Is $|I|$ comfortably above s ? A homography with 5 inliers from 4 samples is meaningless. Demand a real margin — 15–20 minimum, more if you can.
- ☐ **Relative support.** Is w plausible for your sensor and scene? A sudden drop from 0.7 to 0.2 is a failure signal, not a hard scene.
- ☐ **Residual distribution.** Inlier residuals should look like a truncated Gaussian. Bimodal or uniform residuals mean multiple structures or a wrong threshold.
- ☐ **Conditioning.** Check the condition number / determinant of the estimated model. Degenerate fits are often near-singular.
- ☐ **Geometric plausibility.** Does the homography preserve orientation? Does the recovered translation have sane magnitude? Are triangulated points in front of both cameras?
- ☐ **Spatial spread of inliers.** Inliers clustered in one image corner indicate a local structure, not a global model.
- ☐ **Stability.** Run it 10 times with different seeds. If the answers disagree materially, the problem is under-constrained and no single answer is trustworthy.

Part VIII · The variant zoo, organised

Forty years of papers. But recall §3.2: RANSAC has exactly five stages, and **every variant is a better idea about one of them**. That turns the zoo into a table.



9.1 · The variants that matter, by stage

Stage 1 — Sampling. Uniform sampling wastes almost every draw when w is low.

Variant	Year	The idea
NAPSAC	2002	Draw the first point at random, the rest from its spatial neighbourhood. Inliers of one structure are spatially coherent, so local samples are cleaner. Excellent for multi-structure data; suffers when a structure is small.
PROSAC	2005	Sort correspondences by match quality (e.g. SIFT ratio), then draw from a progressively growing prefix. Starts optimistic, degrades gracefully to uniform. Often 100× fewer samples in practice, with the same worst-case guarantee.
Progressive NAPSAC	2019	Grows the neighbourhood progressively, merging local and global sampling. Combines with PROSAC ordering.
NG-RANSAC	2019	A CNN predicts a sampling weight per correspondence; RANSAC samples from the learned distribution.
NeFSAC	2022	A tiny network <i>rejects</i> minimal samples that are motion-inconsistent or poorly conditioned, before the expensive solve. ~1 order of magnitude speed-up.

Stage 2 — Scoring. "Count the inliers" is the crudest possible quality function.

Variant	Year	The idea
MSAC	2000	Truncated quadratic cost instead of a count. Same speed; indistinguishable at a well-set threshold, ~3× more accurate at a badly-set one (§5.4). OpenCV's USAC default.
MLESAC	2000	Maximise the likelihood under an explicit mixture of Gaussian inliers and uniform outliers, estimating the mixing parameter by EM.
MAGSAC	2019	σ-consensus : rather than fixing a threshold, <i>marginalise the model quality over a range of noise scales</i> . Eliminates the threshold parameter.
MAGSAC+ +	2020	Replaces MAGSAC's quality and polishing steps with iteratively re-weighted least squares, weights from the σ -marginalisation. Often an order of magnitude faster than MAGSAC and more accurate. The default recommendation today.

Stage 3 — Verification. $n \cdot N$ residual evaluations is the bottleneck (§4.9).

Variant	Year	The idea
R-RANSAC with $T_{\{d,d\}}$	2002	Pre-test d random points (usually $d = 1$); reject the hypothesis immediately if any fails.
Randomised RANSAC with SPRT	2005	Wald's sequential probability ratio test: evaluate residuals one by one, stop when the decision is statistically settled.
Optimal Randomised RANSAC	2008	The TPAMI treatment proving SPRT-based verification optimal in expected time.
Preemptive RANSAC	2003	Fixed hypothesis pool, breadth-first scoring with pruning. Bounded, real-time-safe runtime.

Stage 4 — Local optimisation. The single biggest accuracy win after the basic algorithm.

Variant	Year	The idea
LO-RANSAC	2003	Whenever a new best model appears, run an inner loop: refit on inliers with a shrinking threshold, a few times. Brings the empirical sample count down to near the theoretical <code>N</code> and markedly improves accuracy.
LO+-RANSAC	2012	"Fixing the Locally Optimized RANSAC" — fixes the cost and threshold schedule of the inner loop.
GC-RANSAC	2018	Runs graph-cut in the local optimisation step, using spatial coherence: neighbouring points are encouraged to share an inlier/outlier label. This is <code>cv2.USAC_ACCURATE</code> .

Stage 5 — Degeneracy. See §8.1 for the 12× measured effect.

Variant	Year	The idea
DEGENSAC	2005	"Two-View Geometry Estimation Unaffected by a Dominant Plane." Detects plane-dominated support and uses plane-and-parallax to recover the correct <code>F</code> .
QDEGSAC	2006	Handles <i>quasi</i> -degenerate data by estimating a hierarchy of constrained models.

Unified frameworks. The practical culmination.

Framework	Year	The idea
USAC	2013	A modular architecture — sampling, verification, local optimisation, degeneracy — showing the components compose. The basis of OpenCV's <code>USAC_*</code> flags.
VSAC	2021	Introduces <i>independent inliers</i> for near-error-free rejection of wrong models, and runs local optimisation on average only once. Reported as two orders of magnitude faster than MAGSAC++ at the same accuracy — 1–2 ms on a CPU — and it never failed on four standard datasets.

Learned and differentiable. For end-to-end pipelines.

Variant	Year	The idea
DSAC	2017	Makes hypothesis <i>selection</i> differentiable by replacing the arg-max with a probabilistic soft selection, so RANSAC can sit inside a network trained by backpropagation.
NG-RANSAC	2019	Learns <i>where to sample</i> (Stage 1).
∇-RANSAC	2023	Generalised differentiable RANSAC — makes the whole randomised pipeline differentiable, including the sampler, via Gumbel-softmax.

9.2 · What should you actually use in 2026?

Situation	Use
Robust linear regression, tabular data	<code>sklearn.linear_model.RANSACRegressor</code> with an explicit <code>residual_threshold</code>
A geometric primitive in 2-D/3-D	<code>skimage.measure.ransac</code> , or the §5.1 engine with MSAC scoring
Homography	<code>cv2.findHomography(..., cv2.USAC_MAGSAC)</code>
Fundamental matrix	<code>cv2.findFundamentalMat(..., cv2.USAC_MAGSAC)</code> — never the default flag (§8.1)
Essential matrix	<code>cv2.findEssentialMat(..., method=cv2.USAC_MAGSAC)</code>
Matches with a usable quality score	<code>cv2.USAC_PROSAC</code> — and remember to pre-sort the input
Unknown or varying noise level	MAGSAC++ (<code>cv2.USAC_MAGSAC</code> , or the <code>pymagsac</code> package)
Hard real-time budget	<code>cv2.USAC_FAST</code> or Preemptive RANSAC, with a hard iteration cap
Multi-structure data	sequential RANSAC, or Progressive-X
Point-cloud primitives	Open3D <code>segment_plane</code> , or PCL <code>SACSegmentation</code> with a constrained model
Inside a trained network	∇-RANSAC or DSAC

If you remember one line from Part VIII: change `cv2.RANSAC` to `cv2.USAC_MAGSAC`. It is a drop-in replacement, it is more accurate, it is less threshold-sensitive, and on `findFundamentalMat` it is dramatically more robust to degeneracy.

Part IX · Neighbours and alternatives

RANSAC is one point in a larger design space. An expert knows when to leave it.

Method	Breakdown point	Needs a threshold?	Deterministic?	Multiple models?	Best when
Least squares	0 %	no	yes	no	no outliers at all
Hough transform	very high	bin size instead	yes	yes, naturally	low-dimensional parameter space (2-3), many instances
LMedS	50 % (hard ceiling)	no	no	no	≤50 % outliers, noise scale unknown
M-estimators / IRLS	0 % (high-leverage points)	tuning constant	yes	no	mild contamination, good initialisation available
Theil-Sen	≈29 %	no	yes	no	robust simple regression, small <code>n</code>
RANSAC	>50 %, budget-limited	yes	no	no (see §10.1)	high outlier rates, any <code>s</code> , known noise scale
MAGSAC++	>50 %	no (marginalised)	effectively yes	no	as RANSAC, noise scale unknown or varying
Branch-and-bound consensus maximisation	>50 %	yes	yes, globally optimal	no	small problems needing a <i>certificate</i>
GNC / TEASER++	>90 % reported	yes	yes	no	point-cloud registration with extreme outlier rates

Hough transform — RANSAC's older sibling

Both are robust voting schemes, but they vote in opposite directions. The Hough transform (Duda & Hart, 1972) has every **data point vote for all models** compatible with it, accumulating in a discretised parameter space; RANSAC has every **hypothesised model count its supporting points**. Consequences:

- Hough naturally finds **many** instances at once — its accumulator has multiple peaks. RANSAC finds one.
- Hough's cost is exponential in the number of parameters (an accumulator for a 4-parameter model is already painful; for an 8-parameter homography, hopeless). RANSAC's cost is independent of the parameter count except through `s`.
- Hough's accuracy is limited by bin size; RANSAC's by the polish step.

Use Hough for lines and circles in images. Use RANSAC for anything with more than about three parameters.

LMedS — the threshold-free classic

Least Median of Squares (Rousseeuw, 1984) minimises the *median* of squared residuals instead of their sum. It needs no inlier threshold, which is genuinely attractive. Two hard limits: its breakdown point is exactly **50 %** — at 51 % outliers the median residual is an outlier residual and the method fails by construction — and its statistical efficiency under Gaussian noise is poor, so it should always be followed by a refit. It is available as `cv2.LMEDS` and `pcl::SAC_LMEDS`. §8.1 shows it degrading by up to 15× under degeneracy, so it is not a free lunch.

Modern practice: if you want threshold-freedom, use **MAGSAC++** rather than LMedS. It has no 50 % ceiling and it is more accurate.

M-estimators and IRLS — the complement, not the competitor

Iteratively re-weighted least squares with a robust loss (Huber, Cauchy, Tukey) down-weights large residuals rather than excluding them. It is smooth, fast, differentiable — and it needs a starting point in the right basin, because the objective is non-convex. It has **no breakdown-point guarantee** against high-leverage outliers.

The right mental model is that these are **complementary stages, not rivals**:

RANSAC finds the basin; IRLS polishes inside it.

This is precisely the architecture of every serious pipeline. COLMAP runs RANSAC per image pair and then bundle adjustment with a Cauchy loss. MAGSAC++ is literally RANSAC with IRLS as its polish step. If you are choosing between them, you have misread the problem: use both, in that order.

Part X · Advanced topics

10.1 · Multiple models: sequential RANSAC

The simplest multi-model strategy is greedy: fit, remove the inliers, repeat on the remainder.

```
def sequential_ransac(data, fit, residuals, min_samples, threshold,
                     max_models=10, min_support=20, **kw):
    """Greedily extract multiple structures: fit, remove inliers, repeat."""
    data = np.asarray(data)
    remaining = np.arange(len(data))
    models = []
    while len(remaining) >= max(min_samples, min_support) \
        and len(models) < max_models:
        r = ransac(data[remaining], fit, residuals,
                   min_samples, threshold, **kw)
        if r.model is None or r.inliers.sum() < min_support:
            break
        models.append((r.model, remaining[r.inliers]))
        remaining = remaining[~r.inliers]
    return models, remaining
```

Verified on three lines (90 points each, $\sigma = 0.15$) plus 60 uniform clutter points, $t = 0.5$:

```
found 3 structures; 48 points unexplained (60 are true clutter)
model 0: slope=+1.002 intercept=+5.006 support=104 purity=0.865
model 1: slope=-0.992 intercept=-5.004 support= 96 purity=0.865
model 2: slope=+0.152 intercept=+0.003 support= 82 purity=0.951
true lines: (+1.000, +5.000) (-1.000, -5.000) (+0.150, +0.000)
```

All three recovered to three decimal places. Note what the purity figures tell you honestly: each consensus set absorbed roughly a dozen clutter points that happened to lie near its line. That is unavoidable for any threshold-based method and is why 48 rather than 60 points remain.

Known weaknesses of the greedy approach, and they are real:

- **Order dependence.** A point near two structures is assigned to whichever is found first.
- **Error propagation.** A slightly wrong first model steals points and corrupts the second.
- **`min_support` is now a critical parameter.** Too low and you "find" structures in the clutter; too high and you miss small real ones.
- **No global consistency.** Nothing prevents two returned models from being near-duplicates.

For serious multi-model fitting, use a method that optimises all models jointly — **Progressive-X** (Barath & Matas, 2019) or energy-minimisation-based multi-model fitting — rather than this loop. But test the loop first; it is often enough.

10.2 · Preconditioning: normalise, always

For projective models, the design matrix is catastrophically ill-conditioned if you feed it raw pixel coordinates. Entries scale like x^2 ($\approx 10^6$ for a 1000-pixel image) alongside entries of order 1, and the SVD's small singular values disappear into floating-point noise.

Hartley normalisation (Hartley, 1997) is the fix, and it is not optional: translate points so their centroid is at the origin, scale so the mean distance from the origin is $\sqrt{2}$ (2-D) or $\sqrt{3}$ (3-D), estimate, then undo the transformation:

```
 $\hat{x} = T x, \quad \hat{x}' = T' x', \quad \text{estimate } \hat{H} \text{ from the hatted points,} \quad H = T'^{-1} \hat{H} T$ 
```

OpenCV does this internally; if you write your own 8-point or 4-point solver, you must do it yourself. Its omission is the single most common reason a hand-written homography or fundamental matrix solver "almost works".

10.3 · Guided sampling from match quality

If you have per-datum quality scores — SIFT ratio-test values, learned matcher confidences — you are throwing away information by sampling uniformly. PROSAC is the principled route, but a crude version costs three lines and helps immediately:

```
# Sort correspondences best-first, then let USAC_PROSAC exploit the order.
order = np.argsort([m.distance / n.distance for m, n in knn_pairs])
src, dst = src[order], dst[order]
H, mask = cv2.findHomography(src, dst, cv2.USAC_PROSAC,
                             ransacReprojThreshold=3.0,
                             maxIters=10_000, confidence=0.9999)
```

USAC_PROSAC silently assumes the input is already sorted best-first. Passing unsorted data does not error — it just performs like ordinary RANSAC, or worse. This is an easy and invisible mistake.

10.4 · RANSAC inside a larger system

System	Where RANSAC sits
Structure from Motion (COLMAP)	per image pair: F/E/H estimation and geometric verification; then incremental PnP; then global bundle adjustment
Visual SLAM (ORB-SLAM)	relocalisation via PnP-RANSAC; loop-closure verification
Panorama stitching	pairwise homography, plus the inlier count as the <i>match-confidence</i> score deciding which images belong in the panorama
LiDAR odometry	ground-plane extraction; ICP correspondence rejection
Object pose estimation	PnP-RANSAC on 3-D-2-D keypoint correspondences
Point-cloud registration	RANSAC over FPFH feature correspondences for coarse alignment, then ICP

Notice the recurring pattern: **RANSAC is the gate, not the answer**. It produces an initial estimate plus an inlier set, and something smoother and more accurate — bundle adjustment, ICP, IRLS — takes over from there. Designing it into a system means deciding what happens when the gate says no.

Part XI · The tuning playbook

A one-page reference.

Set these four things, in this order

#	Parameter	How to choose
1	<code>s</code> — minimal sample size	<i>Not</i> a choice: it is a property of your model (§7.1). But choosing a <i>better-constrained model</i> reduces it, and that is the highest-leverage decision available.
2	<code>t</code> — inlier threshold	$t \approx 2-3 \sigma$ where σ is your measurement noise. Formally $t = \sigma \sqrt{\chi^2_m(0.95)}$ for codimension m (§4.8). Images: 1-3 px. Depth: your datasheet. Unknown: use MAGSAC++.
3	<code>p</code> — confidence	0.99 to 0.9999. It is logarithmically cheap; do not economise.
4	<code>max_trials</code>	The hard cap protecting you from a low <code>w</code> . Set from §4.3 for the <i>worst</i> <code>w</code> you will tolerate, then let adaptive termination finish early.

Always do these five things

1. **Enable adaptive termination** (§4.6) — with a hard cap.
2. **Polish on the full consensus set** (§3.2, step 5) — a measured 3.3× of free accuracy.
3. **Use MSAC scoring** rather than a raw count (§5.4) — ~3× better when `t` is off.
4. **Raise `w` before tuning anything**: ratio test, mutual nearest neighbour, priors (§7.2, §8.3).
5. **Validate the output** against the §8.6 checklist. `p` is a budget, not a certificate (§4.7).

Never do these five things

1. **Never use a sample larger than minimal**. It feels safer; it is exponentially worse.
2. **Never hard-code `N = 1000`** and hope.
3. **Never trust a large inlier count as proof of correctness** — see the bridging model in §8.2.
4. **Never use `cv2.RANSAC` for `findFundamentalMat`** — 12× worse under degeneracy (§8.1).
5. **Never skip normalisation** in a hand-written projective solver (§10.2).

The 60-second diagnostic

```
Result is unstable between runs
-> t too small, or w too low, or the problem is under-constrained.
    Check |I| first: is it barely above s?

Result has many inliers but is visibly wrong
-> t too large (bridging, §8.2), or degeneracy (§8.1).
    Plot the inliers. Are they one structure, or two?

Result is accurate but too slow
-> do not touch RANSAC. Raise w (better matching) or lower s
    (better-constrained model). 445x measured, §7.2.

Result works on dataset A, fails on dataset B
-> t is tuned to A's noise level. Switch to MAGSAC++.

Result is fine offline, jittery in a video stream
-> non-determinism (§8.5). Seed from the previous frame's inliers,
    or only re-estimate when support drops.
```

Part XII · Exercises

Work these in order. Solutions follow; the code from Part IV is all you need.

1 (warm-up). Without running anything: you are fitting a homography ($s = 4$) to matches that are 65 % correct. How many trials for 99 % confidence? For 99.99 %?

2 (the cost of non-minimal samples). Modify the engine to sample 6 points instead of 4 for a homography. Predict the change in N at $w = 0.5$ from §4.3, then measure it. Why does the accuracy *not* improve?

3 (threshold sensitivity). Reproduce Panel C of Figure 1 for the *circle* model. Does the $2\text{--}3\sigma$ rule still hold? Should it?

4 (breakdown point). Push the line problem's outlier ratio from 40 % up to 95 % in steps, with `max_trials = 1,000,000`. Find where RANSAC breaks. Is the failure gradual or abrupt?

5 (degeneracy). Generate 200 points on a *line* in 3-D, add 50 off-line points, and fit a **plane**. What happens, and why does the `sv[1]` test in `fit_plane` matter so much here?

6 (scoring). Implement MLESAC's scoring rule (an EM estimate of the inlier mixing proportion, maximising the mixture log-likelihood) and compare it to MSAC on the §5.4 loose-threshold experiment.

7 (verification). Implement the $T_{\{1,1\}}$ pre-test: before computing all n residuals, test one random point and reject the hypothesis if it fails. Measure the reduction in residual evaluations at $w = 0.3$. Does the total trial count change? Should it?

8 (real data). Take two photographs of the same building from different positions. Run the §7.3 pipeline with `cv2.RANSAC` and with `cv2.USAC_MAGSAC`, on both `findHomography` and `findFundamentalMat`. Explain the difference you see in the F case using §8.1.

Solutions

1. $N = \lceil \log(0.01) / \log(1 - 0.65^4) \rceil$. Since $0.65^4 = 0.1785$, $N = \lceil \log(0.01) / \log(0.8215) \rceil = \lceil 23.4 \rceil = \text{**24**}$. For $p = 0.9999$: $N = \lceil \log(0.0001) / \log(0.8215) \rceil = \lceil 46.9 \rceil = \text{**47**}$. The un-rounded values are exactly double ($23.42 \rightarrow 46.85$), as §4.2 property 2 promises — confidence is logarithmically cheap.

2. At $w = 0.5$, s goes $4 \rightarrow 6$, so N goes $72 \rightarrow 293$: **4× more trials**. Accuracy does not improve because the *polish* step already fits all ~ 100 inliers; the minimal sample's only job is to be uncontaminated. A 6-point sample is a strictly worse lottery ticket for the same prize. This is the intuition behind "always minimal" and the single most common beginner error.

3. Yes, and it must — the derivation in §4.8 depends only on the residual being a 1-codimensional Gaussian distance, which is true for radial distance to a circle just as for perpendicular distance to a line. What *does* change is the degeneracy rate: collinear triples are common in a circle sample, so more trials are wasted, and you will see the trial count rise.

4. The failure is **abrupt**, not gradual — this is the two-message lesson of §7.2. Accuracy holds almost unchanged until the trial budget can no longer find a clean sample, then collapses completely. With $s = 2$, $w = 0.05$ still needs only 1,841 trials, so a line survives to extreme contamination; the binding constraint becomes whether 5 % of the data still *determines* a line better than some accidental alignment of the outliers. Expect a cliff somewhere in the low single digits of w , and expect its exact location to depend on n .

5. This exercise has a deliberately uncomfortable answer, and it is the most important one in the set. Every 3-point sample drawn entirely from the line is collinear, hence degenerate, and that is the *common* case since the line holds 200 of the 250 points. The `sv[1]` test correctly rejects all of those — so you might expect the algorithm to report failure. It does not.

Measured over 20 runs, RANSAC confidently returns a plane every single time, with **202-207 of 250 points as inliers**. The reason is that any sample of two line points plus one off-line point is *not* collinear, passes the test, and defines a perfectly valid plane — the one containing the line and that third point. Every such plane holds the whole line, so every one of them scores ~200. The tell is in the variation across runs: the recovered normal has a standard deviation of `[0.23, 0.68, 0.46]` across its components, yet in every run it is orthogonal to the line direction to within `|n·d| < 0.001`. The plane's *rotation about the line* is completely unconstrained, and RANSAC picks one at random.

The lesson: a sample-level degeneracy test cannot detect data-level degeneracy. No amount of checking the minimal sample tells you that the data as a whole under-determine the model. Only the §8.6 validation checklist does — here, by noticing that repeated runs disagree wildly while all reporting high support. This is the same failure as the dominant plane in §8.1, in its purest form.

6. MLESAC will land between MSAC and plain counting, closer to MSAC. The reason is instructive: MLESAC's extra machinery estimates the mixing proportion γ , which mostly matters when the outlier distribution is genuinely uniform over a known region. Its cost per hypothesis is higher (EM iterations), so on this problem MSAC wins on accuracy-per-millisecond. This is why MSAC, not MLESAC, became OpenCV's default.

7. Residual evaluations drop by roughly the fraction of hypotheses that are bad — at `w = 0.3` with `s = 2`, about 91 % of hypotheses are contaminated, so you should see close to an order of magnitude fewer evaluations. The **trial count is unchanged**, and it must be: `T_{1,1}` changes only the cost of *verifying* a hypothesis, not the probability of *drawing* a good one. Strictly, it introduces a small chance of rejecting a good hypothesis whose one test point happened to be an outlier, so the required trial count rises very slightly — the analysis of that trade-off is exactly the content of the SPRT papers.

8. For `findHomography` on a planar façade the two flags will agree closely — a façade is genuinely a plane, so the homography model is correct and well-constrained. For `findFundamentalMat` on the same pair, `cv2.RANSAC` will return a poor `F` because the façade is a dominant plane and the epipolar geometry is under-determined by planar points (§8.1); the off-façade points — a lamp post, a tree, a passing car — are what actually constrain it, and there are few. `USAC_MAGSAC`'s degeneracy handling detects this and recovers. You will see the difference most clearly by drawing epipolar lines for the off-plane points.

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Glossary

Term	Meaning
All-inlier / uncontaminated sample	A minimal sample containing no outliers. The event whose probability is w^s .
Breakdown point	The smallest fraction of arbitrarily corrupted data that can make an estimate arbitrarily wrong.
Codimension	The number of independent directions in which a datum can deviate from the model. Sets the χ^2 degrees of freedom for the threshold (§4.8).
Consensus set	The points whose residual to a hypothesis is below the threshold.
Degenerate sample	A sample from which the model is not uniquely determined (collinear points for a plane, coplanar ones for $\mathbb{F}$).
Gross error	An outlier: a measurement that is not a noisy observation of the model at all.
Hypothesise-and-verify	The generate-then-test architecture RANSAC belongs to.
Inlier ratio w	The fraction of data that are inliers. The single most important quantity in every cost formula.
Local optimisation	An inner refinement loop run when a new best model appears (LO-RANSAC, GC-RANSAC).
Minimal sample / s	The fewest data needed to determine the model. Always use exactly this many.
Pseudo-outlier	An outlier w.r.t. the model of interest that belongs to a <i>different real structure</i> (§8.2).
Sampson distance	A first-order approximation to geometric reprojection error; the standard residual for $\mathbb{F}$ and $\mathbb{E}$.
σ-consensus	MAGSAC's technique of marginalising model quality over a range of noise scales instead of fixing a threshold.
SPRT	Wald's sequential probability ratio test, used to abandon hopeless hypotheses early.
Total least squares	Fitting that minimises perpendicular distance, appropriate when all coordinates are noisy (§5.2).
$T_{\{d,d\}}$ test	Verify d random points first; reject immediately if any fails.

The whole tutorial in eight sentences **RANSAC** = **RAN**dom **SA**mple **C**onsensus, Fischler & Bolles, 1981. It is a *meta-algorithm*: you supply a minimal solver, a residual function and a threshold. It repeatedly fits a *minimal* sample and keeps the model with the best-supported *consensus set*. Always use the minimal sample size — smaller samples are exponentially more likely to be clean. $N = \log(1-p) / \log(1-w^s)$ trials suffice — remarkably, the dataset size n barely enters. That budget guarantees a clean sample was *drawn*, not that the answer is *right* — so validate. The threshold is the one parameter that matters: use $2-3\sigma$, or MAGSAC++ if you cannot know σ . Then always refit on all the inliers — and if it is too slow, raise w , don't tune RANSAC.